

When the Treatment Talks Back: Information Limits of Observational Logs from Adaptive Generative Political Communication*

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Abstract

Large language models now conduct persuasive political conversations that adapt turn by turn to the human they are persuading. We formalize the treatment delivered by such a system as a composite object — a strategy policy π , a generation kernel G , and an information environment E — and define deployment, policy, and turn-level excursion estimands for it. We then show that conversation logs collected under a deployed policy carry structurally limited information about turn-level causal effects. First, when the deployed policy has history-deterministic components, turn-level effects are nonparametrically unidentified (support failure); in RLHF-tuned deployments this condition is structurally frequent and design-reinforced. Second, even with formal positivity, the semiparametric efficiency bound for population excursion effects degrades exponentially in the policy’s effective inverse temperature: no regular estimator in the nonparametric class escapes the rate (Result A); under a uniform-recurrence condition the bound compounds geometrically across turns (Proposition A-T). Third, the overlap weight mass concentrates on the policy’s learned decision boundary as temperature falls, so the causal information that remains stably observable is localized there; the boundary effect is a distinct — and policy-relevant — estimand β^{ov} , not a silent replacement for the population effect (Result B, proved in one dimension). A phase diagram in (n, τ) with an exact boundary — $\log \mathbb{E}[1/p_\tau] = \log(1 + \tau \sinh(1/\tau))$ for our main design — separates estimable, variance-explosion, and deceptively stable-but-biased regimes. Re-querying three RLHF-tuned open-weights models on 200 real debate contexts shows median per-context strategy entropy of exactly zero and modal-strategy shares of 94.5–98.6%, implying $\geq 10^5$ conversations for any minority-strategy contrast. An audit of 56,831 deployed persuasion conversations shows that randomized prompt instructions explain only 12–32% of realized evidence-dose variance. We propose the generative micro-randomized trial (G-MRT) — within-conversation turn-level randomization with a locked evidence bank, gated generation, and assignment-ITT analysis — as the design remedy.

1 Introduction

Conversational AI persuades. In randomized trials, GPT-4 debates shifted opponents’ positions more often than human debaters did (Salvi et al., 2025), and conversations with frontier models

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moved policy attitudes with durable effects (Hackenburg et al., 2025). Yet the meta-analytic picture of political persuasion is one of small and heterogeneous effects ($g \approx 0.02$, $I^2 \approx 76\%$), which makes decomposition — *which turn-level behaviors of the AI cause belief change* — the scientifically and politically urgent question. The strongest lever in Hackenburg et al. (2025), a reward model selecting responses turn by turn, is precisely the component whose interior remains a black box: it is an adaptive turn-level policy, and its logs are what an analyst inherits.

This paper asks what those logs can and cannot identify. Our answer is pessimistic in a precise way and constructive in a precise way.

Pessimistic: the treatment in these systems is not a message but a composite (π, G, E) — a strategy policy, a generation kernel, and an information environment. Turn-level contrasts computed from deployment logs face three distinct failures: (i) support failure where the policy is history-deterministic (Proposition 1); (ii) an exponential collapse of the semiparametric efficiency bound in the policy’s decisiveness even when positivity formally holds (Result A), with bounds that transfer a fortiori from single turns to conversations and compound geometrically across turns (Proposition A-T); and (iii) localization — the overlap mass that survives concentrates on the policy’s decision boundary, so what remains stably estimable is a boundary estimand β^{ov} , distinct from the population effect (Result B). We verify each mechanism in simulation, locate real deployments on the resulting (n, τ) phase diagram using re-query experiments on three RLHF-tuned models, and audit instruction–behavior slippage in 56,831 deployed conversations.

Constructive: each failure corresponds to a specific design element of a generative micro-randomized trial (G-MRT): uniform turn-level randomization restores support and overlap; exogenous assignment removes sequential confounding; regime-defined estimands resolve ill-defined treatments; and a fidelity protocol with assignment-ITT analysis handles generated-treatment noncompliance.

Contributions, three and locked: (1) an estimand framework for generative treatments with a general adaptive action space; (2) temperature-indexed overlap collapse and localization — Proposition 1, Result A with explicit conditions, Proposition A-T, Result B proved in one dimension, the fixed- τ counterexample, and the (n, τ) phase diagram; (3) the G-MRT design with a generated-treatment fidelity protocol. Everything else — generator transport, internal-representation methods, simultaneous inference, constrained policy learning — is explicitly out of scope (Section 9).

2 Related Work

Texts as treatments, static. Imai and Nakamura (2026) identify effects of generated texts as fixed objects; ordering effects *within* a fixed object involve no human-feedback loop and no positivity problem. Our object is different: an adaptive policy operating *on* human responses, where the policy itself destroys overlap. This distinction — dynamic response policy versus static generated object — is the boundary line of this paper.

Off-policy evaluation with deficient support. The estimation-side face of our problem is studied in the contextual-bandit and reinforcement-learning literatures: doubly robust off-policy evaluation (Dudík et al., 2011; Thomas and Brunskill, 2016; Jiang and Li, 2016), and, closest to Proposition 1, off-policy learning under *deficient support*, where the logging policy assigns zero or near-zero probability to some actions (Sachdeva et al., 2020). We draw the boundary in four places. First, that literature works at the level of specific estimators and their variance

or regret; Results A–B are statements about the *semiparametric efficiency bound* — no regular estimator in the nonparametric class escapes the rate, which closes the “use a better estimator” escape by construction. Second, we index the collapse by an interpretable policy parameter (effective temperature of an RLHF-shaped softmax) and obtain exact boundary geometry in (n, τ) , including a closed form for the phase boundary in our main design. Third, localization (Result B) — *where* the surviving information lives, and the $\tau \downarrow 0$ limit of the overlap-weighted estimand — has, to our knowledge, no counterpart in that literature, whose remedies (truncation, regularized weights, support restriction) implicitly change the estimand without characterizing the limit object. Fourth, our remedy is a *design* (randomize the turns), not an estimator: the setting is scientific inference about turn-level persuasion effects, where the analyst can run an experiment, not a fixed logged dataset to be salvaged.

Micro-randomized trials. MRTs randomize pre-defined discrete actions at decision points and target causal excursion effects via WCLS (Qian et al., 2022; Boruvka et al., 2018). We import the excursion estimand and add what conversation requires: a generation kernel between assignment and realization, ill-defined observational labels, and fidelity auditing of generated treatments. LLM-in-MRT precedents exist in behavior-change messaging; our novelty claim is confined to within-conversation turn-level randomization of generative political persuasion with information locking.

Stochastic interventions. Regime-specific potential outcomes for stochastic and modified policies (Díaz Muñoz and van der Laan, 2012; Kennedy, 2019; Young et al., 2014) supply our formal object: β is defined under a kernel intervention — assign the label, realize from G . Our addition is the history-dependent generation kernel and the decisiveness collapse.

Overlap weighting. β^{ov} connects to overlap weights and empirical equipoise (Li et al., 2018); Result B gives the $\tau \downarrow 0$ limit of that estimand under a softmax policy, and Section 4 argues the limit object is itself policy-relevant.

Information equivalence. Dafoe et al. (2018) motivate our separation of form from dose: a persuasive-strategy manipulation that also changes information content is confounded in exactly the sense our fidelity audit quantifies.

Support failure is old; its systematicity is new. Nonidentification from deterministic treatment assignment is standard since Robins (1986). Our claim is not the structure but its origin: RLHF preference optimization widens score gaps and deployment favors low temperature, so history-determinism in deployed LLM policies is *structurally frequent and design-reinforced* — it cannot be escaped by collecting better logs from the same deployment.

3 Generative Treatments

Individuals $i = 1, \dots, n$ converse over turns $t = 1, \dots, T$. At turn t the system draws an action from its policy, $A_{it} \sim \pi_t(\cdot \mid H_{it})$, where H_{it} is the interaction history, and realizes an utterance from its generation kernel, $M_{it} \sim G_\theta(\cdot \mid A_{it}, H_{it}, E_j)$, where E_j is the information environment (for us: a locked evidence bank on issue j). The user responds R_{it} ; outcomes Y_i include belief accuracy, calibration, reactance, and one-week retention.

The action space is general. $A_t \in \mathcal{A}$ may be a discrete strategy label, a continuous generation parameter, or a factorial cell. As a running example — and as the implementation object of the companion experimental paper — we use a 2×2 : form $F \in \{\text{assertive}, \text{interrogative}\} \times \text{dose}$ $D \in \{\text{high: 4–5 facts}, \text{low: 1–2 facts}\}$.

Treatment is a stochastic intervention. Potential outcomes are defined under regimes, $Y_i^{\pi, G}$, with do-notation read as a kernel intervention: intervene on the label, realize from G (Díaz Muñoz and van der Laan, 2012; Young et al., 2014). Three estimand layers:

- **Deployment effect** $\Delta^{\text{dep}} = \mathbb{E}[Y^{\pi_1, G_1}] - \mathbb{E}[Y^{\pi_0, G_0}]$: what existing RCTs identify.
- **Policy effect** $\Delta^{\text{pol}}(G) = \mathbb{E}[Y^{\pi_1, G}] - \mathbb{E}[Y^{\pi_0, G}]$: holding the generator fixed.
- **Turn-level excursion effects** (primary): contrasts of one turn’s action against a pre-specified reference policy ρ (uniform over $\mathcal{A}$, all t , all h), with subsequent turns following ρ . Proximal excursions (immediate reactance, belief update) are standard CEE/WCLS targets (Boruvka et al., 2018); distal excursions (final accuracy, retention) connect to distal causal excursion effect estimation (Shi and Dempsey, 2025) — using ordinary WCLS for terminal outcomes is a known attack point we avoid by construction.

Ill-defined treatment in logs. An observational label $\tilde{S}_t = s$ corresponds to the realized kernel $M_t \sim G(\cdot \mid s, H_t)$, a *different distribution at every history*. Label-pooled contrasts do not correspond to any well-defined treatment pair. Our regime definition does not “solve” this nonidentification; it replaces the question with a well-posed one — and the well-posed β is identified only under assignment randomization (Section 5).

4 Why Transcripts Are Not Enough

Model the deployed policy’s action choice as softmax in a history-dependent score gap with effective temperature τ : for the two-action case, $p_\tau(h) = \sigma(\Delta(h)/\tau)$ with σ the logistic function. Deployment practice pushes τ down; preference optimization pushes $|\Delta|$ up.

4.1 Support failure

Proposition 1 (Support failure). *If $\pi_t^{\text{dep}}(a \mid h) = 0$ on a positive-probability set of histories, turn-level excursion effects are nonparametrically unidentified.*

The counterexample pair requires no latent confounder — the failure is support, not confounding. Take $T = 1$, $H \in \{h_0, h_1\}$, $\pi^{\text{dep}}(1 \mid h_0) = 1/2$, $\pi^{\text{dep}}(1 \mid h_1) = 0$, $Y(a) \mid H = h \sim N(\mu_a(h), 1)$, and let two DGPs differ only in $\mu_1(h_1) = c_1$ versus c_2 . The observed-data law factorizes as $P(H) \pi^{\text{dep}}(A \mid H) N(y; \mu_A(H), 1)$, to which the $(1, h_1)$ stratum contributes nothing; the two DGPs are observationally identical yet their uniform-reference excursion effects differ by $P(h_1)(c_2 - c_1)$. The multi-turn version replaces H with $(X, R_{1:t-1})$ and recovers the time-varying-confounding form. The structure is Robins (1986); the LLM-specific claim is frequency and reinforcement, not inevitability.

4.2 Exponential collapse of the information bound

With $\tau > 0$, positivity holds formally, so the right response to Proposition 1 alone would be “use IPW.” Result A closes that escape at the level of the efficiency bound.

Proposition 2 (Result A). *Assume (A1) the high-gap set $\mathcal{H}_\delta = \{h : |\Delta(h)| \geq \delta\}$ has positive measure; (A2) conditional outcome variances are bounded below by $\underline{\sigma}^2$ on $\mathcal{H}_\delta$; (A3) the model class is nonparametric in the outcome regression — no parametric extrapolation into unobserved strata. Then the semiparametric efficiency bound for the ATE-type excursion satisfies*

$$V_\tau \geq \underline{\sigma}^2 \cdot \mathbb{P}(\mathcal{H}_\delta) \cdot e^{\delta/\tau}.$$

Proof. Insert $\min(p_\tau, 1 - p_\tau) \leq e^{-\delta/\tau}$ on $\mathcal{H}_\delta$ into the bound of Hahn (1998), $V_\tau = \mathbb{E}[\sigma_1^2/p_\tau + \sigma_0^2/(1 - p_\tau) + (\beta(H) - \beta)^2]$, discarding the third term and restricting to $\mathcal{H}_\delta$. $\square$

By the convolution theorem, *no regular estimator in the class* attains lower asymptotic variance: switching from IPW to AIPW recovers no counterfactual information that the sample does not contain. Parametric extrapolation escapes the bound, but that is buying identification with an assumption, and should be said in exactly those words. Effective sample size is arm-specific Kish: $n_{\text{eff}}(a) = n/\mathbb{E}[1/\pi_\tau(a | H)]$. The slogan “halving τ squares the required n ” holds within fixed-gap strata; with heterogeneous gaps the max-gap stratum dominates the rate.

Remark 1 (Single-turn bounds are a fortiori multi-turn bounds). Our results are stated for one turn and two actions, while the motivating setting is a T -turn conversation. The gap runs in the harmless direction. The turn- t excursion estimand conditions on the observed history H_t and contrasts actions at that turn, with subsequent turns following ρ . The observed-data model factorizes turn by turn; the turn- t factor is exactly the single-turn problem with propensity $\pi_\tau(\cdot | H_t)$, and identifying β_t additionally requires bridging every later turn from π^{dep} to ρ . Discarding the information cost of those later-turn bridges can only *understate* the difficulty; hence any lower bound for the single-turn problem at the (Δ, τ) prevailing at turn t is also a lower bound for the conversational problem. (K -action policies reduce to the two-action case via the two arms of the contrast.)

Proposition 3 (A-T: multi-turn compounding). *Consider the T -turn policy value $V(\rho)$ of the uniform reference ρ over K actions, observed under π_τ . Start from the semiparametric efficiency bound for finite-horizon off-policy evaluation with history-dependent policies (Kallus and Uehara, 2020; Jiang and Li, 2016):*

$$V_{\text{eff}} = \text{Var}(V_1(H_1)) + \sum_{t=1}^T \mathbb{E}_\pi \left[\left(\prod_{s \leq t} w_s \right)^2 \text{Var}(V_{t+1} | H_t, A_t) \right], \quad w_s = \frac{\rho(A_s | H_s)}{\pi_\tau(A_s | H_s)}.$$

Assume (M1) uniform recurrence of the high-gap set: $\mathbb{P}(H_{s+1} \in \mathcal{H}_\delta | h, a) \geq p_\delta > 0$ for every $h \in \mathcal{H}_\delta$ and every action a ; and (M2) a per-turn conditional variance floor $\underline{\sigma}^2$ on $\mathcal{H}_\delta$. Then

$$V_{\text{eff}} \geq \underline{\sigma}^2 \cdot \mathbb{P}(H_1 \in \mathcal{H}_\delta) \cdot p_\delta^{T-1} \cdot \left[\frac{e^{\delta/\tau}}{K^2} \right]^T,$$

which compounds geometrically in T whenever $\tau < \delta/\log(K^2/p_\delta)$.

Proof sketch. All terms of V_{eff} are nonnegative; keep the terminal one. On $\mathcal{H}_\delta$, $\mathbb{E}_\pi[w_t^2 \mid H_t = h] = K^{-2} \sum_a 1/\pi_\tau(a \mid h) \geq K^{-2} e^{\delta/\tau}$ since the rarest action has probability at most $e^{-\delta/\tau}$. Induct backward from $t = T$: by the tower property and (M1) — whose action-uniformity is what lets the bound pass through each turn’s weight — each turn contributes a factor $p_\delta e^{\delta/\tau}/K^2$, and turn 1 contributes $\mathbb{P}(H_1 \in \mathcal{H}_\delta) e^{\delta/\tau}/K^2$. The self-contained derivation of the quoted efficiency bound is deferred to the appendix; our contribution is the insertion and the induction. $\square$

Substantively, (M1) says decisiveness persists regardless of what the system does — as with strongly reactant partisan users. The K^2 factor reflects the conservativeness of the uniform reference and is not optimized; the excursion version (turn t , reference thereafter) follows with exponent $T - t + 1$. The honest reading: more turns make the information limit exponentially worse, now as a proposition rather than a heuristic.

4.3 Localization, the counterexample, and the phase diagram

Proposition 4 (Result B, one-dimensional). *Let $\omega_\tau(H) = p_\tau(1 - p_\tau)$ and $\beta_\tau^{\text{ov}} = \mathbb{E}[\omega_\tau\{Y(1) - Y(0)\}]/\mathbb{E}[\omega_\tau]$. Assume H has continuous density f ; $\Delta \in C^1$ with isolated simple roots; a tail-separation condition ($\inf_{|h-r| \geq \epsilon} |\Delta(h)| > 0$ for all $\epsilon > 0$); and continuity of f and of $c(h) = \mathbb{E}[Y(1) - Y(0) \mid H = h]$ near the roots, c bounded. Then, as $\tau \downarrow 0$, the normalized overlap measure converges weakly to a point mass at the decision boundary; with a unique root at 0,*

$$\mathbb{E}[\omega_\tau] = \tau \frac{f(0)}{|\Delta'(0)|} (1 + o(1)), \quad \beta_\tau^{\text{ov}} \rightarrow \mathbb{E}[Y(1) - Y(0) \mid H = 0].$$

Proof idea. $\omega_\tau = K(\Delta(h)/\tau)$ exactly, with $K(u) = 1/[4 \cosh^2(u/2)]$ and $\int K = 1$: an approximate-identity kernel. The tail-separation condition kills contributions away from the roots at rate $e^{-c(\epsilon)/\tau}$; near a simple root, the change of variables $v = \Delta(h)$, $v = \tau u$ and dominated convergence give $\int g \omega_\tau f dh = \tau g(0)f(0)/|\Delta'(0)| + o(\tau)$ for bounded continuous g . Ratios deliver the three claims. With multiple roots, mass distributes proportionally to $f(r_j)/|\Delta'(r_j)|$. The multivariate coarea version remains at appendix/conjecture status. $\square$

Why the boundary estimand matters substantively. β^{ov} is not a consolation prize; it is the effect in a policy-relevant population. The decision boundary $\{\Delta(H) = 0\}$ collects the histories at which the deployed policy is genuinely undecided — the algorithmic analogue of clinical equipoise (Li et al., 2018). Three consequences. First, *marginal policy evaluation*: any small change to the policy — a regulation, a guardrail, a reward-model update, a temperature change — alters behavior precisely on and near this boundary, so the effect among boundary histories is the first-order term in the value change of any local policy modification. Second, *auditability*: β^{ov} is the one turn-level causal quantity a regulator could estimate from logs alone with stable precision, which makes its scope and its limits worth stating exactly. Third, the political reading, stated as interpretation rather than result: histories far from the boundary — strongly reactant partisan contexts where the policy is most decisive — are exactly where population-level causal information vanishes; what deployment data can tell us about persuasion is confined to the persuadable margin as *the policy* defines it, not as the scientist would.

What Result B does *not* say. At fixed $\tau > 0$, an IPW estimator with true propensities targets the *population* effect as $n \rightarrow \infty$ — the estimand does not silently become the boundary effect because the policy is sharp. Numerically: $\tau = 0.3$, $n = 4 \times 10^6$, $\beta(H) = 1 + |H|$ gives

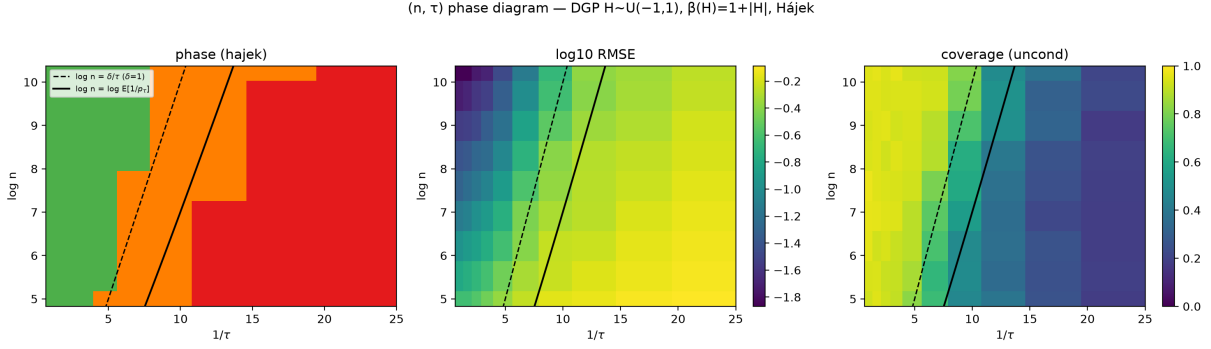


Figure 1: (n, τ) phase diagram; DGP $H \sim U(-1, 1)$, $\beta(H) = 1 + |H|$, Hájek estimator, 400 replications per cell. Left: regime classification (green estimable, orange variance explosion, red biased-stable). Middle, right: $\log_{10}$ RMSE and unconditional coverage. Dashed line: fixed-gap heuristic $\log n = \delta/\tau$. Solid line: exact boundary $\log n = \log \mathbb{E}[1/p_\tau] = \log(1 + \tau \sinh(1/\tau))$. Regime thresholds (relative bias 0.10/0.20, coverage 0.85) are presentation choices defended by the RMSE and coverage panels.

IPW 1.792 $\approx$ ATE 1.798 $\neq \beta^{\text{ov}} = 1.356$. Formal identification and practical estimability are different properties; our claims are about the *rate* of information loss and the *location* of surviving information, never about a change of target.

The joint limit is the real object. The failure surface lives in (n, τ) : with $n \gg e^{\delta/\tau}$ the population effect is estimable (at exploding variance); near $n \sim e^{\delta/\tau}$ variance explodes; with $n \ll e^{\delta/\tau}$ rare crossover actions never appear in the sample and estimators become *biased while looking stable* — the deceptive regime. Figure 1 shows the three regimes for the main design $H \sim U(-1, 1)$, $\Delta(H) = H$, with two boundary lines: the fixed-gap heuristic $\log n = \delta/\tau$ (dashed) and the exact $\log n = \log \mathbb{E}[1/p_\tau] = \log(1 + \tau \sinh(1/\tau))$ (solid; Monte-Carlo agreement to four digits; $\tau \downarrow 0$ asymptote $1/\tau + \log \tau - \log 2$). The empirical regime boundary tracks the solid line.

4.4 Where real deployments sit: Study 0 (descriptive, motivating)

Three empirical bridges locate deployed systems on the diagram. Their epistemic status is *motivating*: they estimate proxies of policy decisiveness and instruction–behavior slippage, not the parameters of Sections 4.1–4.3.

(0-a) Reproduction. Our pipeline reproduces the direction of [Salvi et al. \(2025\)](#) on the released data ($n = 750$): personalized Human-AI has the largest crude odds ratio for increased agreement ($2.18 > 1$), matching the original ordering.

(0-b) Re-query decisiveness. For 200 rebuttal contexts stratified by treatment and prior-agreement strength, we regenerate $k = 20$ responses per context and classify the persuasion strategy of each. Across three RLHF-tuned open-weights families the per-context strategy distribution is near-degenerate (Table 1).

Substituting per-strategy $\hat{p}_s$ into the effective-sample formula (Dirichlet-smoothed, $k = 20$ resolution): the modal strategy needs $\sim 3 \times 10^3$ conversations per contrast arm, every minority

model	median entropy (bits)	zero-entropy contexts	modal share
Qwen2.5-7B-Instruct	0.0	89.5%	REBT 98.6%
Mistral-7B-Instruct-v0.3	0.0	58.0%	REBT 94.5%
Phi-3.5-mini-instruct	0.0	75.5%	REBT 97.3%

Table 1: Re-query decisiveness across model families: 200 contexts $\times k = 20$ regenerations each; strategies classified by a single 32B judge. Low entropy recurs across families — the observable signature of “structurally frequent.”

strategy $\geq \sim 1.1 \times 10^5$ (a lower bound — unobserved strategies may be rarer than smoothing can register), and a two-minority contrast needs the *sum* of two such terms. These are illustrative insertions into Result A’s formula, not estimates of any deployment’s τ .

Placing contexts on the phase diagram. Only the composite Δ/τ — the score gap in temperature units, exactly the phase diagram’s abscissa — is identified from choice frequencies; τ alone is not. Estimating the two-action reduction $\widehat{\Delta}/\tau = \log[(c_1 + \alpha)/(c_2 + \alpha)]$ per context yields a median of 3.71 for all three models — which is the *resolution ceiling* of $k = 20$ draws: 58–89.5% of contexts are right-censored there (the runner-up action was never realized). The reflexive reading: measuring a deployed policy’s decisiveness runs into the same information limit that blocks effect estimation, so all decisiveness figures from finite re-query are lower bounds; via Proposition A-T the per-turn stratum factor $\mathbb{E}[1/p] \geq 42$ compounds across turns.

Two robustness results discipline these numbers. *Judge sensitivity:* relabeling all 4,000 Qwen responses with a 32B judge yields raw agreement 0.903 but $\kappa = 0.13$, and $\kappa \approx 0$ once both-modal pairs are excluded — judges agree on the dominant label and disagree almost completely on minority-label identity. Aggregate quantities are nevertheless stable (per-strategy n_{req} ratios 0.88–1.52 across judges). We therefore report entropy, modal share, and n_{req} — and we do not interpret minority-label identity before human double-coding (pre-registered $\kappa \geq 0.6$ on 200 turns). *Label-error simulation:* propagating misclassification rates $q \in \{0.05, 0.1, 0.2\}$ through the pipeline (symmetric and majority-absorbing noise models anchored to the observed confusion pattern) moves rare-strategy n_{req} by -25% to $+6\%$ even at $q = 0.2$; symmetric noise biases toward optimism, so the 10^5 scale is conservative.

(0-c) Instruction $\neq$ behavior. In the released data of Hackenburg et al. (2025) (56,831 persuasion conversations across three studies), the randomized rhetoric instruction explains $\eta^2 = 0.20/0.12/0.32$ of realized fact-count variance in studies 1/2/3 — 68–88% of realized-dose variation occurs *within* assignment. Assignments also bundle: “information” realizes 8.3–21.8 facts on average while “moral reframing” realizes 1.6–2.1, and 52–56% of moral-reframing conversations realize zero facts. Where compliance flags exist, valence compliance is 72.7% (study 1). What was randomized was an instruction; the treatment that reached the participant was a generated behavior, substantially decoupled from it. This is the observational counterpart of the fidelity problem the G-MRT protocol addresses by design.

5 The G-MRT Design as the Remedy

Each failure in Section 4 maps to a design element:

failure	design element
support failure + bound collapse	uniform turn-level randomization $\rho(a h) = 1/4 \geq \epsilon$
sequential confounding	exogenous assignment of A_t
ill-defined treatment	regime-defined (kernel-intervention) estimands
generated-treatment noncompliance	assignment-ITT + fidelity protocol

The protocol: (1) conversation-level randomization of the generator $Z_i \in \{G_a, G_b\}$, at least one open-weights; (2) turn-level uniform randomization of A_t with full logging of assignment, probability, and realization; (3) a locked evidence bank of ≥ 20 verified fact units per issue with per-turn sub-banks or without-replacement sampling *specified as part of the treatment regime* — dose is honestly named a high-information communication package (fact count covaries with length and cognitive load; length-controlled designs are a follow-up); (4) a fidelity gate that inspects each utterance before sending, regenerates once on mismatch, and sends-with-log on second failure. The gate is part of the treatment kernel: ITT identifies the effect of *assignment to the generate-and-gate protocol $\tilde{G}$* , and the paper says so in those words. Differential noncompliance across cells (e.g., interrogative form failing more often) contaminates cell contrasts; we pre-register a fidelity floor of $\geq 80\%$ per cell with demotion of affected contrasts, publish gate-decision logs, and pre-specify a no-gate sensitivity analysis, since a gate classifier whose errors correlate with assignment would itself distort the treatment.

The price of identification (illustrative, not a theorem). Mixing exploration into deployment, $\rho = (1 - \epsilon)\pi_\tau + \epsilon \text{Unif}$, caps the variance contribution at $\sigma^2 K / (n\epsilon)$ at a per-turn regret cost scaling with ϵ — a design corollary quantifying the trade-off, presented as an illustration because its exact form depends on outcome bounds, estimator, and horizon.

6 Estimation

Proximal excursions: WCLS with known randomization ρ (Boruvka et al., 2018; Qian et al., 2022). Distal excursions: DCEE estimators (Shi and Dempsey, 2025), not terminal-outcome WCLS. Policy values: cross-fitted doubly robust estimators for 2–3 pre-specified policy pairs only. Scope sentence: all guarantees are relative to the G-MRT’s known ρ ; nothing in this section rescues transcript-only analysis. Multiplicity: corrections for a pre-specified family, finalized after pilot; simultaneous confidence regions are companion-paper material.

7 Simulation Evidence

Four executed studies; theory-to-DGP correspondence is declared object-by-object in the repository.

(i) Sign reversal under adaptivity. A defensiveness-adaptive policy makes the naive turn-level contrast wrong in *sign* (truth +0.068; naive −0.294, coverage 0.0) while known- ρ IPW/DR recover it (bias ≤ 0.01 , coverage ≈ 0.95 –0.97) — the confounding face of the problem, and the G-MRT’s licence (Figure 2).

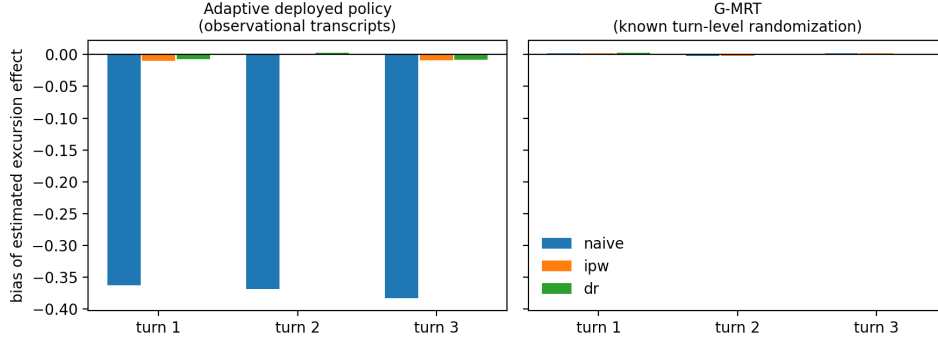


Figure 2: Sign reversal under an adaptive policy: naive turn-level contrasts versus known- ρ IPW/DR.

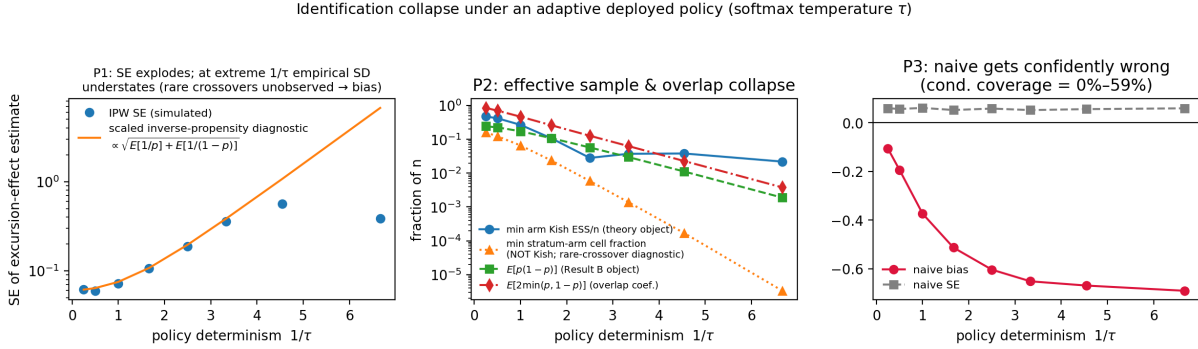


Figure 3: Two-state temperature collapse: predictions P1–P3.

(ii) **Two-state temperature collapse.** Along a τ grid calibrated so $\tau = 1$ reproduces the adaptive policy of (i): IPW SE grows exponentially, tracking a scaled inverse-propensity diagnostic curve (labeled as such — it is not the efficiency bound); the naive estimator becomes *more confident while more wrong* as τ falls (coverage $\rightarrow 0$ with shrinking SE) (Figure 3). Metric discipline: global arm-specific Kish ESS (theory object), minimum stratum-arm cell occupancy (rare-crossover diagnostic — not Kish), $\mathbb{E}[p(1-p)]$ (Result B object), and $\mathbb{E}[2\min(p, 1-p)]$ (diagnostic) are reported as four separate columns, coverage both conditional and unconditional on estimability, with zero-cell rates alongside. One finding we flag for practice: in the collapse regime the *realized* Kish ESS rebounds non-monotonically ($0.028 \rightarrow 0.53$) exactly when the zero-stratum-arm rate jumps ($0.70 \rightarrow 0.99$) — surviving weights homogenize as rare crossovers vanish from the sample, so a healthy-looking realized ESS co-occurring with high zero-cell rates is itself the signature of the deceptive regime. ESS should never be reported without zero-cell rates.

(iii) **Continuous-boundary battery.** ($H \sim U(-1, 1)$ main, $N(0, 1)$ robustness; $\beta(H) \in \{1, 1 + e^{-H^2}, 1 + |H|\}$; 400 replications per cell.) At $n = 32,000$: all population-targeting estimators are unbiased with nominal coverage down to $\tau \approx 0.15$; at $\tau = 0.045$ Horvitz–Thompson has SD 2.95 (variance explosion) while Hájek shows bias 0.46 with SD 0.30 — biased-yet-stable; oracle AIPW does not rescue coverage (Result A is not an IPW artifact); the oracle outcome-

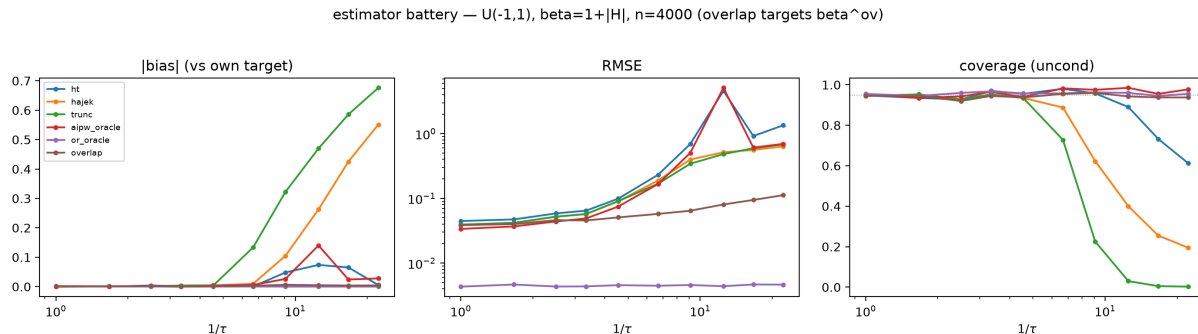


Figure 4: Estimator battery on the continuous-boundary design at $n = 4,000$ (bias against each estimator’s own target; the overlap estimator targets β^{ov}).

regression line is flat by construction — the (A3) escape made visible, identification bought by assumption; and the overlap estimator remains unbiased with nominal coverage *for its own target* β^{ov} at every τ , with $\beta^{\text{ov}} \rightarrow \beta(0) = 1$ as Result B predicts (Figure 4). Under $N(0, 1)$ the collapse is earlier and, at extreme τ , propensities underflow to machine 0/1 so IPW becomes numerically undefined — reported as estimability 0, not dropped.

(iv) **Phase diagram.** The empirical regime classification over the (n, τ) grid tracks the exact solid boundary $\log \mathbb{E}[1/p_\tau]$; the dashed δ/τ heuristic is visibly off at small τ (Figure 1).

8 Empirical Plan and Ethics

The companion experimental paper implements the 2×2 G-MRT: pilot $n = 80$ –120 (function verification: assignment mechanics, fidelity rates, bank leakage, dropout — not effect detection), demonstration $n = 180$ –250 with pre-registered minority contrasts, confirmatory sample from post-pilot power simulation. Pre-registration locks the fidelity floor, blocking factors (issue), attrition handling (IPW-for-attrition), and reactivity checks (randomizing measurement itself in pilot). Ethics: persuasion toward *accuracy* on factual political questions with a fact-checked locked bank (cf. Costello et al., 2024); no deception arms; disclosure of AI interlocutor; IRB before pilot. The three Study 0 analyses involve only public data and no new human subjects.

9 Discussion

What we rule in: estimand definitions that survive the generative setting; explicit information limits with rates and locations; a design that restores identification at a quantified exploration cost. What we rule out of this paper: generator transport (effects across model versions — motivated empirically by cross-model heterogeneity in Chen et al., 2026), internal-representation adjustment (overlap problems of representation conditioning are themselves design-based cautionary tales, cf. Tierney et al., 2025), simultaneous inference over excursion families, and epistemically constrained policy learning. Each is a companion, not a section.

The one-sentence version of this paper: *adaptive systems generate the least population-level causal information exactly where they are most decisive, and the information that survives lives*

on their decision boundary — so if you want turn-level causal knowledge from conversational AI, you must randomize the turns.

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